Subhransu S. Bhattacharjee

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AI PhD candidate at ANU developing generative, multimodal, and evaluation systems for decision-making under uncertainty. First-author CVPR 2025 and ECCV 2026 research on generative world models and spatial reasoning, with applied-science depth in LLM and RAG evaluation at Microsoft and quantitative machine learning at Optiver and JPMorgan, and maintained open-source research software.

Education

Doctor of Philosophy **Artificial Intelligence**
School of Computing, Australian National University, Australia *Apr 2023 – Mar 2027 (expected)*
- **Thesis topic:** *Into the Unknown: Tackling Spatio-semantic Uncertainty using Generative Posteriors*
- **Supervisors:** Dr. Rahul Shome, Dr. Dylan Campbell, and Prof. Stephen Gould

Bachelor of Engineering (Honours) **Mechatronics**
School of Engineering, Australian National University, Australia *Jul 2018 – Dec 2022*
- **First Class Honours**; GPA 6.48/7.00; ranked 3rd in cohort. **Minors:** Mathematics; Electronic Communication Systems.
- **Machine Learning**, London School of Economics Summer School — **Grade A**.
- **Thesis project:** Whiplash Gradient Descent Dynamics.

Industry Research Experience

Microsoft India Development Center **Jun 2026 – Present**
Applied Scientist PhD Research Intern, LLM-as-Judge, RAG & Agent Evaluation
- Built the backbone of an end-to-end LLM-as-judge evaluation framework for RAG and agent-generated outputs, combining claim-level groundedness, relevance-purity, response-discipline, and dissatisfaction-aware evaluation around an existing proprietary system.
- Developed an agentic ingestion harness that converts unstructured, Likert-scored annotations into structured evaluation records, reducing annotator cognitive load by $\sim$35% and increasing usable annotation throughput $\sim$2$\times$, supporting golden-set construction, evaluator calibration, and failure analysis.

Optiver, Sydney, Australia **Nov 2024 – Feb 2025**
Quantitative Research Intern, Neural–Fuzzy Modelling, xLSTM, XGBoost, SHAP
- Developed a neural–fuzzy joint classifier for Korean retail-flow prediction, achieving **94% classification accuracy**, integrating xLSTM and XGBoost within a streaming market-feature pipeline with online SHAP attribution.
- Built leakage-aware walk-forward evaluation and model-monitoring infrastructure for Korean market microstructure, and separately developed and backtested a statistical-arbitrage model for the Hong Kong market.

JPMorgan Chase & Co., Sydney, Australia **Dec 2021 – Feb 2022**
Quantitative Research Intern, NLP, Information Retrieval, Risk
- Developed a metadata-indexing and retrieval pipeline over 10,000+ financial documents supporting transformer-based question answering and structured answer extraction for risk-research workflows, improving top-k retrieval recall by $\sim$8 percentage points.

Selected Publications

Subhransu S. Bhattacharjee, Dylan Campbell & Rahul Shome: *Believing is Seeing: Unobserved Object Detection using Generative Models. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. Project page.
Introduces unobserved object detection with generative priors that infer occluded and out-of-frame objects across 2D, 2.5D, and 3D scenes.

Subhransu S. Bhattacharjee, Dylan Campbell & Rahul Shome: *FlatLands: Generative Floormap Completion From a Single Egocentric View. European Conference on Computer Vision (ECCV)*, 2026. Springer. Paper.
Releases FlatLands: 270,575 observations from 17,656 real metric indoor scenes across six datasets.
Compares 11 methods, showing stochastic generators improve completion fidelity and capture layout uncertainty.

Subhransu S. Bhattacharjee, Hao Lu, Dylan Campbell & Rahul Shome: *MatterDoor: Sampling Zero-shot Spatio-semantic Priors using Generative Models. arXiv:2510.11014*, 2025. Under review; RSS 2026 FM4RoboPlan workshop presentation, Sydney. Preprint.
Samples environment priors from partial observations using pretrained generative models.
Produces RGB-D point-cloud hypotheses for occupancy, target semantics, and configuration-space planning.

Subhransu S. Bhattacharjee & Ian Petersen: *Analysis of the Whiplash Gradient Descent Dynamics. Asian Journal of Control* (Special Issue), Wiley, 2023. DOI: 10.1002/asjc.3153.
A feedback-based inertial gradient method with adaptive damping achieves provable polynomial and exponential convergence rates that extend classical optimisation theory to high-dimensional settings.

Subhransu S. Bhattacharjee & Ian Petersen: *A Closed Loop Gradient Descent Algorithm Applied to Rosenbrock's Function. Australian & New Zealand Control Conference (ANZCC)*, 2021. DOI: 10.1109/ANZCC53563.2021.9628258.
Introduces adaptive closed-loop damping for inertial gradient descent on a stiff non-convex optimisation benchmark.

Open-Source Research & Software

TorchKAN — Independent PyTorch implementation of Kolmogorov–Arnold Networks with spline, Legendre (KAL-Net), Chebyshev (KAC-Net), and convolutional (KANvolver) variants; classification and regression experiments, CUDA execution, post-training quantisation, and Integrated-Gradients interpretability. Sole maintainer. Project Site | Code

Mangroves — Python package for depth-based hierarchical data management: type-aware ingestion, dynamic attribute access, result caching, and CPU–GPU tensor movement for episodic training pipelines. Sole author; maintenance-only. Project Site | Code

TuringTree — Vectorless, reasoning-based RAG running fully on-device (Ollama + Qwen) with 0–100 confidence scoring and abstention; tests, CI, and a tagged release. Co-developed by a four-person team during the Microsoft Global Intern Hackathon. Project Site | Code

Research Experience

Research School of Management, Australian National University **Sep 2023 – Sep 2024**
Graduate Research Assistant — FinTech & AI, PI: Dr Priya Muthukannan
- Developed AI-adoption assessment frameworks for an industry-linked Commonwealth Bank of Australia project, translating open-banking and dynamic-capabilities analysis into strategy recommendations.

CIICADA Lab, Australian National University **Dec 2022 – Feb 2023**
Summer Research Scholar — Control & Optimisation, Supervisor: Prof Ian Petersen
- Advanced feedback-based optimisation analysis for systems without closed-form solutions, contributing to publications at ASCC 2022 and ANZCC 2021.

School of Computing, Australian National University **Mar 2022 – Jun 2022**
Undergraduate Research Scholar — Geometry of Data, Supervisor: Prof Richard Hartley
- Tested invertibility of differentiable neural mappings with dense positional encoding, achieving a 72% hit rate under an RMSE criterion.

DRDO, India **May – Jul 2020**
Summer Research Trainee — Passive Radar Signal Processing, Kalman Filtering, FPGA, Real-Time DSP
- Developed a Kalman-filter-centred method for selecting matched filters for noisy passive-radar signals, and implemented an FPGA-based multiprocessor interface for real-time long-range radar analysis.

Selected Achievements

2026 Research contributor to a Google Research–supported ANU Robotics project on world models for task-and-motion planning.
2026 Selected invitee to the Hugging Face, PyTorch, and Red Hat Meetup, Bengaluru.
2025 Invited Research Talk at the Dyson Robotics Laboratory, Imperial College London
2025 Vice-Chancellor's Travel Grant, Australian National University, for CVPR 2025 travel.
2025 Invited attendee, Citadel & Citadel Securities PhD Summit, London.
2024 Optiver PhD Quant Lab Program (selected participant).
2023 ANU International University Research Scholarship (4 years) and Higher Degree by Research Merit Stipend.
2022 Highly Recommended Paper at the *Asian Control Conference*; Best Student Reviewer Award.
2021 High Commendation Award for undergraduate student paper at the *Australia & New Zealand Control Conference*.

Teaching & Service

Senior/Head Tutor, Australian National University: Introduction to Machine Learning; Building Cyber-Physical Systems; Responsible Leadership; Power Systems & Control Theory — delivering ML mathematics, optimisation, robotics, and microprocessor labs.
Reviewing: CVPR; ECCV; NeurIPS; ICML; AAAI; ICRA; IROS; IEEE RA-L; Asian Journal of Control (2021–2026).

Technical Skills

Programming & Scientific Computing: Python; PyTorch; SQL; C; MATLAB; Bash; NumPy; SciPy; pandas; scikit-learn.
Generative, Multimodal & Spatial AI: diffusion; flow matching; transformers; uncertainty estimation; 3D vision; OpenCV; Open3D; PyTorch3D; COLMAP; ROS2.
LLM Evaluation, NLP & Retrieval: LLM-as-judge evaluation; RAG evaluation; human-preference evaluation; information retrieval; transformer QA; document indexing; agent evaluation.
Quantitative Methods: probability; statistics; optimisation; market microstructure; temporal validation; backtesting; XGBoost; xLSTM; SHAP.
Research Systems: Linux; Git; Docker; Apptainer; SLURM; CUDA; Weights & Biases.